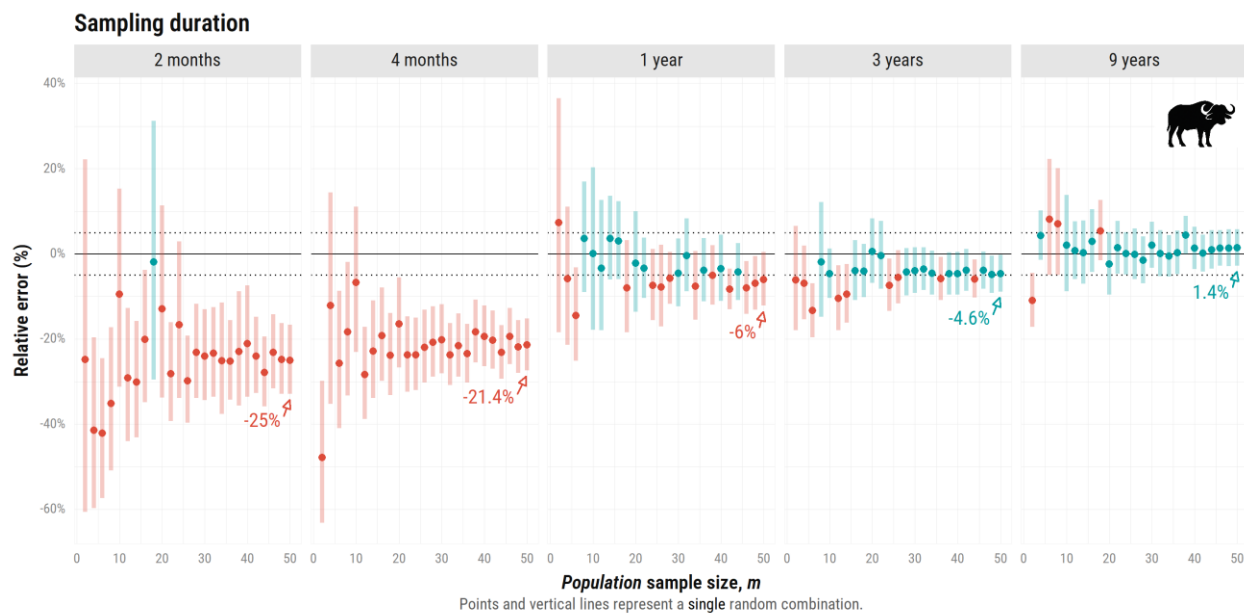


# Too few, too many, or just right? Optimizing sample sizes for population-level inferences in animal tracking projects

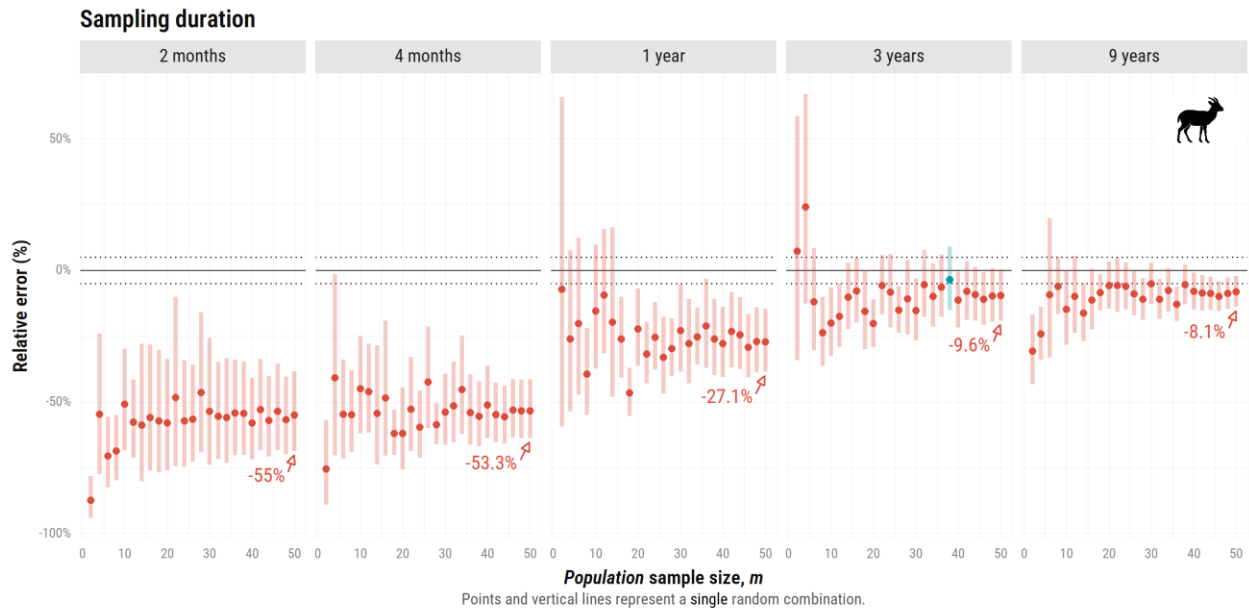
## S3. Single combination plots

### Home range estimation



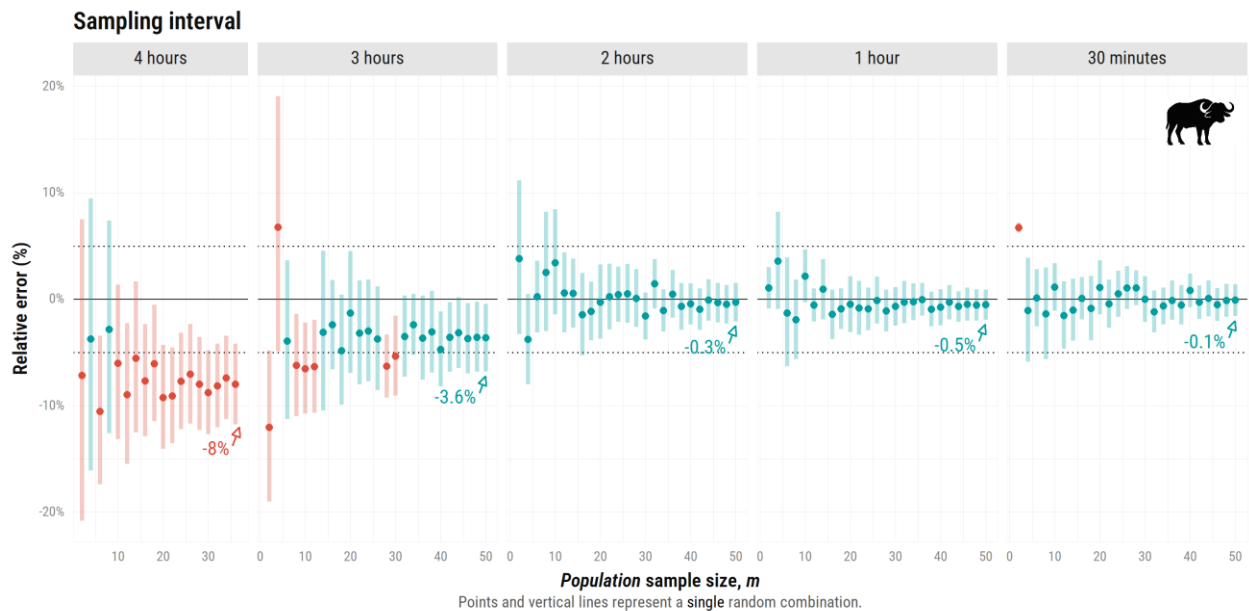
**Figure S3.1.** Relative error (%) in home range estimates (AKDE) as a function of population sample size ( $m$ ), ranging from 2 to 50 simulated African buffalos (*Syncerus caffer*). Each vertical facet corresponds to one of five sampling durations: 2 months, 4 months, 1 year, 3 years, and 9 years. The

plot shows mean estimates based on a **single** combination of individuals per  $m$  (no resampling); estimates in **blue** are within the error threshold and in **red** if estimates fall outside of it. Horizontal dotted lines mark the  $\pm 5\%$  error threshold.



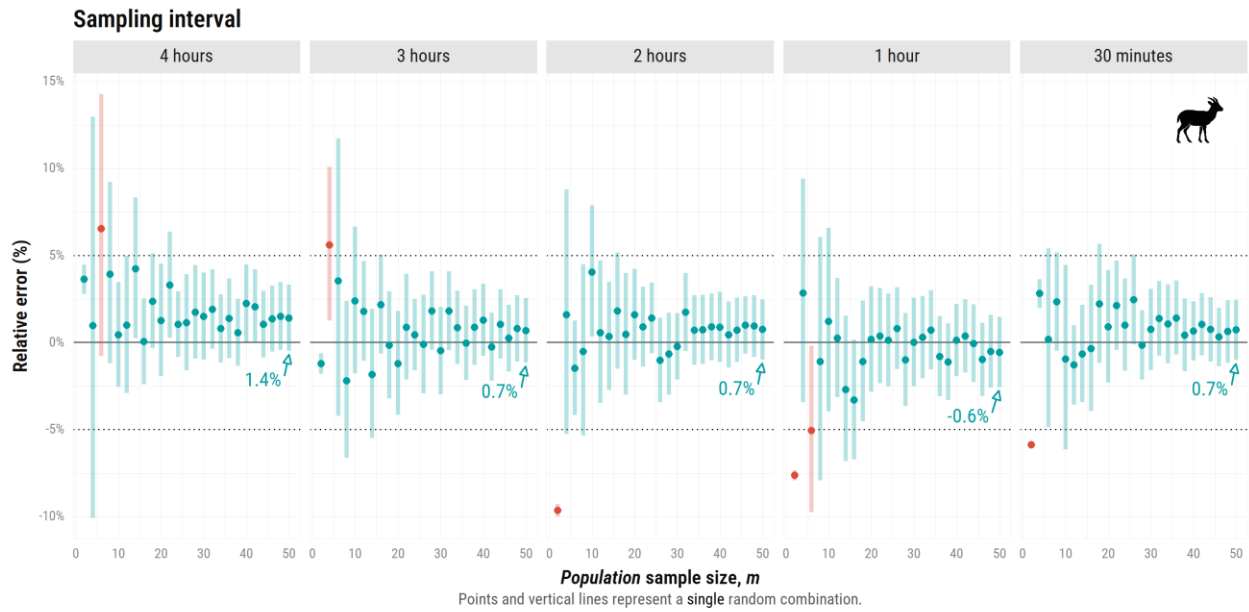
**Figure S3.2.** Relative error (%) in home range estimates (AKDE) as a function of *population sample size* ( $m$ ), ranging from 2 to 50 simulated Mongolian gazelles (*Procapra gutturosa*). Each vertical facet corresponds to one of five sampling durations: 2 months, 4 months, 1 year, 3 years, and 9 years. The plot shows mean estimates based on a **single** combination of individuals per  $m$  (no resampling); estimates in **blue** are within the error threshold and in **red** if estimates fall outside of it. Horizontal dotted lines mark the  $\pm 5\%$  error threshold.

## Speed & distance estimation



**Figure S3.3.** Relative error (%) in speed estimates (CTSD) as a function of *population sample size* ( $m$ ), ranging from 2 to 50 simulated African buffalos (*Syncerus caffer*). Each vertical facet corresponds to one of five sampling intervals: 4 hours, 3 hours, 2 hours, 1 hour, and 30 minutes. The

plot shows mean estimates based on a single combination of individuals per  $m$  (no resampling); estimates in blue are within the error threshold and in red if estimates fall outside of it. Horizontal dotted lines mark the  $\pm 5\%$  error threshold.



**Figure S3.4.** Relative error (%) in speed estimates (CTSD) as a function of *population sample size* ( $m$ ), ranging from 2 to 50 simulated Mongolian Gazelles (*Procapra gutturosa*). Each vertical facet corresponds to one of five sampling intervals: 4 hours, 3 hours, 2 hours, 1 hour, and 30 minutes. The plot shows mean estimates based on a **single** combination of individuals per  $m$  (no resampling); estimates in blue are within the error threshold and in red if estimates fall outside of it. Horizontal dotted lines mark the  $\pm 5\%$  error threshold.